Charles “Chaz” Merritt

Curriculum Vitae

Athens, GA · charles.merritt@uga.edu · LinkedIn · GitHub · Website

Research Interests

Markov Processes, Computational Forestry, Embeddings, Machine Learning

Education

M.S. in Artificial Intelligence 2026 — GPA: 3.66
University of Georgia, Institute for Artificial Intelligence
Thesis: *Stochastic Valuation of Planted Loblolly Pine Stands Using Monte Carlo Simulation*
Advisor: Pete Bettinger

B.A. in Cognitive Science 2024 — GPA: 3.53
University of Georgia, Institute for Artificial Intelligence
Minor in Computer Science

Research Experience

Research Professional May – July 2026
University of Georgia, Warnell School of Forestry & Natural Resources
Developed a landscape scale timber modeling and optimization framework. Developed an ArcGIS Python toolbox for semi-automated feature extraction.

Research Assistant Aug. 2024 – May 2026
University of Georgia, Institute for Artificial Intelligence
Developed stochastic simulation models to incorporate uncertainty into stand-level forest management decisions. Lead the full-stack development of a web interface for a complex timber supply optimization model. Contributed to peer-reviewed research at the intersection of ML and forestry.

Other Experience

Geospatial Analyst Intern May – Sep. 2024
Northwest Management
Designed automated GIS pipelines using Python, Rasterio, and GeoPandas. Implemented clustering algorithms and image segmentation models (SAM) for forest analysis. Processed and analyzed large-scale spatial datasets for operational forestry decision-making.

OMNI Intern May – Dec. 2023
Oak Ridge National Lab
Supported development of ML-based intrusion prevention systems (IPS) for enterprise networks. Gained hands-on experience with SmartNICs (Nvidia BlueField DPUs).

Publications

Peer-Reviewed Articles

Merry, K., Bettinger, P., **Merritt, C.**, Rasheed, K., Lowe III, R. C., & da Silva, B. (2026). AI Technology Use in North American Forestry. *Journal of Forestry.* `doi:10.1007/s44392-026-00079-8`

Presentations

Merritt, C. (2025). Optimizing Timber Stands under Uncertainty. *Society of American Foresters Conference.*

Merritt, C. (2025). Harnessing Reinforcement Learning for Forest Growth Modeling Under Uncertainty. *AI For Eastern US Forestry Conference.*

Merritt, C. (2023). Online Learning in Applied Forestry. *PERSEUS Annual Meeting.*

Merritt, C. (2023). Rapid & Scalable Forest Inventory with Deep Neural Nets. *Southern Forestry & GIS Conference.*

Merritt, C. (2023). Exploring AI Frameworks for Next-Generation Cybersecurity with Nvidia Morpheus & SLIPS. *Dept. of Energy OMNI Fire Poster Session.*

Selected Projects

- **Optimization Software Web Interface** (2026). Developed an interactive webapp for the SETS timber supply models. Tools: Django, Python, TypeScript, HiGHS-Py.
- **Brandon AI** (2026). Developed contrastive representation models for phonetic similarity learning with InfoNCE loss. Tools: PyTorch, NumPy.

Honors and Awards

ISACA Foundation Digital Trust Scholarship — Jan. 2024

Service

- Club Archery @ UGA, *President* — Aug. 2024 – May 2025

Skills

- **Programming:** Python, TypeScript
- **Machine Learning:** PyTorch, Reinforcement Learning, Contrastive Learning, MCMC
- **Modeling:** Stochastic Optimization, Simulation, Value-at-Risk (VaR, CVaR)
- **Data & GIS:** GeoPandas, Spatial Analysis, Clustering, Remote Sensing
- **Tools:** Django, Node.js, HiGHS, NumPy, Railway

Certifications

New Media, University of Georgia — May 2024

Applied Data Science, University of Georgia — May 2024